Modeling ELO: does team strength change over a season?

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Abstract

1 Introduction

2 Methods

2.1 Model 1: constant team strength

$$p_i^{(k)} = \text{expit}(\beta^{(k)} + b_{h_i}^{(k)} - b_{a_i}^{(k)})$$

$$\frac{d\ell}{d\beta} = \sum_{i=1}^{N}(Y_i - p_i)$$

$$\frac{d\ell}{db_j} = \sum_{i:j=h_i}(Y_i - p_i) - \sum_{i:j=a_i}(Y_i - p_i) - b_j/\tau \tag{2.1}$$

Algorithm 1: Alternating estimation algorithm for maximizing $f(Y,b)$

Input: Y, number of iterations K, minimum variance $\varepsilon > 0$, gradient ascent step size $\delta > 0$

Output: Parameter estimates τ, β, and b_j

Initialize $\tau^{(0)} = \varepsilon$, $\beta^{(0)} = 0$, and $b_j^{(0)} = 0$ for all j

for $k = 1$ **to** K **do**

  for $j = 1$ **to** m **do**

    Update $b_j^{(k)}$ by solving $\frac{d\ell}{db_j} = 0$ for b_j after setting $p_i = p_i^{(k-1)}$ and $\tau = \tau^{(k-1)}$ in (2.1).

  end

  Update $\tau^{(k)} = \max\left\{\sum_{j=1}^{m}\left(b_j^{(k-1)}\right)^2/m, \varepsilon\right\}$

  Update $\beta^{(k)} = \beta^{(k-1)} + \delta \cdot \frac{d\ell}{d\beta}\big|_{p^{(k-1)}}$

end

return $\tau^{(K)}$, $\beta^{(K)}$, *and* $b_j^{(K)}$

2.2 Model 2: changing team strength

To allow for team strength/ELO to change over the course of the season, we adjust the previous model to allow the strength of team j (b_j) to become a function of time. Thus we now let $b_{j,t}$ denote the team strength of team j on the tth day of the season (t days after opening day). Our new model is:

$$\text{logit}(\text{pr}(Y_i = 1)) = \beta + b_{h_i,t_i} - b_{a_i,t_i}$$

We still assume the marginal distribution $b_{j,t} \sim N(0,\tau)$ with $b_{j,t}$ independent of $b_{j',t'}$ for any t, t' if $j \neq j'$. However, we should allow for correlation between $b_{j,t}$ and $b_{j,t'}$ that should decay the further apart in time t is from t'. We model this by introducing the correlation parameter $0 \leq \rho \leq 1$ and assuming

$$\text{cov}(b_{j,t}, b_{j,t'}) = \tau\rho^{|t-t'|}$$

- What is the complete-data likelihood $f(Y,b)$ for model 2?

$$f(Y,b) = \left(\prod_{i=1}^{N} p_i^{Y_i}(1-p_i)^{1-Y_i}\right)\left(\prod_{j=1}^{M}(2\pi)^{-N/2}|\Sigma_j|^{-1/2}\exp\left(-\frac{1}{2}b_j^\top\Sigma_j^{-1}b_j\right)\right) \tag{2.2}$$

- Calculate the score equations akin to what we did for model 1 and propose an algorithm to estimate the $b_{j,t}$ (and the model parameters) that maximizes the likelihood. Note that we need only estimate $b_{j,t}$ for values of t for which team j played a game.

$$\ell(Y,b) = \sum_{i=1}^{N} \left[Y_i \log(p_i) + (1-Y_i)\log(1-p_i) \right] - \frac{1}{2} \sum_{j=1}^{M} \left[N_j \log(2\pi) + \log|\Sigma_j| + b_j^\top \Sigma_j^{-1} b_j \right] \tag{2.3}$$

$$\frac{\partial \ell}{\partial \beta} = \sum_{i=1}^{N} (Y_i - p_i) \tag{2.4}$$

$$\frac{\partial \ell}{\partial b_{j,t}} = \sum_{i:j=h_i,t=t_i} (Y_i - p_i) - \sum_{i:j=a_i,t=t_i} (Y_i - p_i) - (\Sigma_j^{-1} b_j)_t \tag{2.5}$$

$$\frac{\partial \ell}{\partial \tau} = -\frac{1}{2} \sum_{j=1}^{M} \left\{ \mathrm{tr}\left(\Sigma_j^{-1} \frac{\partial \Sigma_j}{\partial \tau} \right) - b_j^\top \Sigma_j^{-1} \frac{\partial \Sigma_j}{\partial \tau} \Sigma_j^{-1} b_j \right\} \tag{2.6}$$

$$\frac{\partial \ell}{\partial \rho} = -\frac{1}{2} \sum_{j=1}^{M} \left\{ \mathrm{tr}\left(\Sigma_j^{-1} \frac{\partial \Sigma_j}{\partial \rho} \right) - b_j^\top \Sigma_j^{-1} \frac{\partial \Sigma_j}{\partial \rho} \Sigma_j^{-1} b_j \right\} \tag{2.7}$$

$$\frac{d\Sigma_j}{d\rho} = \begin{bmatrix} 0 & & \Delta t \rho^{\Delta t - 1} \\ & \ddots & \\ \Delta t \rho^{\Delta t - 1} & & 0 \end{bmatrix} \tag{2.8}$$

$$\frac{d\Sigma_j}{d\tau} = \begin{bmatrix} 1 & & \rho^{\Delta t} \\ & \ddots & \\ \rho^{\Delta t} & & 1 \end{bmatrix} \tag{2.9}$$

Algorithm 2: Alternating estimation algorithm for Model 2

Input: Y, number of iterations K, minimum variance $\varepsilon > 0$, gradient ascent step size $\delta > 0$, gradient descent step size $\eta > 0$, gradient ascent step size $\gamma > 0$

Output: Parameter estimates τ, β, ρ, and $b_{j,t}$

Initialize $\tau^{(0)} = \varepsilon$, $\beta^{(0)} = 0$, $\rho^{(0)} = 1$, and $b_{j,t}^{(0)} = 0$ for all j and relevant t

for $k = 1$ **to** K **do**

  for $j = 1$ **to** m **do**

    for $t = 1$ **to** T_j **do**

      Update $b_{j,t}^{(k)}$ by solving $\frac{d\ell}{db_{j,t}} = 0$ for $b_{j,t}$ after setting $p_i = p_i^{(k-1)}$, $\tau = \tau^{(k-1)}$, and $\rho = \rho^{(k-1)}$ in (2.5)

    end

  end

  Update $\tau^{(k)} = \tau^{(k-1)} - \gamma \cdot \frac{d\ell}{d\rho}\big|_{p^{(k-1)}}$

  Update $\beta^{(k)} = \beta^{(k-1)} + \delta \cdot \frac{d\ell}{d\beta}\big|_{p^{(k-1)}}$

  Update $\rho^{(k)} = \rho^{(k-1)} - \eta \cdot \frac{d\ell}{d\rho}\big|_{p^{(k-1)}}$

  Project $\rho^{(k)}$ back onto the interval $[0,1]$ if it falls outside the range

end

return $\tau^{(K)}$, $\beta^{(K)}$, $\rho^{(K)}$, *and* $b_{j,t}^{(K)}$